

KRISHNAN IYER

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SUMMARY

Currently, my research focuses on *in-network computation* and *reconfiguration* support in hardware, specifically FPGAs. This enables users to push on-demand hardware acceleration, offering a superior alternative to limited rigid ASIC acceleration or higher latency on-path CPUs.

EDUCATION

KAUST (King Abdullah University of Science and Technology) Jan 2025 – Dec 2028

Ph.D. in Computer Science

GPA: 3.77/4.0

KAUST (King Abdullah University of Science and Technology) Jan 2023 – Dec 2024

M.S. in Computer Science

GPA: 3.79/4.0

Amrita Vishwa Vidyapeetham

Apr 2016 – Aug 2020

B.Tech. in Electrical and Electronics Engineering.

GPA: 8.61/10

RESEARCH EXPERIENCE

Weave Framework: Semantics-Aware FPGA Abstraction

Jun 2025 – Now

Supervisor: Prof. Suhaib Fahmy

KAUST

- Designed **Weave**, a framework that derives FPGA partial-reconfiguration (PR) design from an application's execution flow, lowering reconfiguration overhead and improving area utilization while hiding PR behind ordinary function calls. Implemented it as a DSL for execution flow, a placement compiler, an automated build tool, and a runtime that manages region state, bitstream selection, and dependency resolution.
- Reduced application response time by up to $11\times$ over a uniform-slot baseline by deriving region placement from the application's execution semantics.
- Increased per-slot resource utilization, packing $2\times$ as many reconfigurable regions onto a **ZCU102** by sizing each region to its function group rather than to the largest function.
- Decreased PR build time $\sim 5\times$ (17 h to 3 h) by automating the toolflow with slot-scoped abstract shells.

OffRAC: Remote Accelerator Calls on FPGAs

Jan 2025 – Now

Supervisor: Prof. Suhaib Fahmy

KAUST

- Extended the benchmarking suite to split each request across a configurable number of packets with server-side reassembly, enabling realistic variable request/response workloads with accurate latency measurement.
- Added FPGA-backed OffRAC benchmarking over the **EasyNet** transport to evaluate end-to-end CPU-to-FPGA offload workloads.
- Implemented hugepage-backed logging buffers for per-thread latency tracing, enabling long-duration measurement at minimal fast-path overhead.
- Ported and tuned the **libtpa**/DPDK userspace TCP stack to the **BlueField-3** DPU (ARM/aarch64), adding build fixes and DPDK SoC configuration.
- Created a hostless **EasyNet** variant, removing PCIe logic and floorplanning the design to open up room for PR regions, then measured the resulting reconfiguration time and area.

Designing a Partially Reconfigurable FPGA NIC

Jul 2024 – Dec 2024

Supervisor: Prof. Suhaib Fahmy

KAUST

- Designed an architecture extending the open-source Corundum FPGA NIC to support **Partial Reconfiguration (PR) over the network**, swapping hardware functions on demand without interrupting connectivity, and validated it on a **Xilinx Alveo U50**.

- Built a reconfiguration controller managing bitstream storage and retrieval over DMA and ICAP, driven by a custom UDP packet format that carries the target region and bitstream.
- Built an initial prototype that also extended **OpenNIC** (*by Xilinx*) to support Partial Reconfiguration.

PUBLICATIONS

- [1] S. Fahmy and K. Iyer, "Dynamic and partial reconfiguration of FPGAs," in *Handbook of Computer Architecture*, A. Chattopadhyay, Ed. Springer Nature, 2024.
- [2] Z. Yang, K. B. Iyer, Y. Chen, *et al.*, "OffRAC: Offloading through remote accelerator calls," 2025. arXiv: 2504.04404 [cs.NI].
- [3] K. B. Iyer, "Designing a partially reconfigurable FPGA NIC," M.S. thesis, KAUST Research Repository, 2024. [Online]. Available: <https://doi.org/10.25781/KAUST-FA7T6>.

HONORS & AWARDS

- **Google Summer of Code '19** – ARM NEON SIMD optimization for dav1d (VideoLAN). 2019
- **Google Summer of Code '18** – SDHCI/MMC driver for Haiku OS. 2018

PROFESSIONAL EXPERIENCE

Space Telelink

May 2022 – Jan 2023

Embedded Researcher

- Implemented multiple device enumeration over a single USB controller (Infineon's FX3), i.e., a composite driver. The driver included HID, mass storage, and a FAT32 file management driver with basic file seek, read, and write functions.
- Explored Ethernet over USB protocols, including RNDIS. This required efficient management of different DMA channels in sync with various USB transfers such as control, interrupt, and bulk transfers. The project led to a fast enumeration of USB peripherals with decent bandwidth, enough to support USB 3.0.
- Implemented a scalable CAN-based communication network for a manipulator robot, resulting in a simplified cable harness and reduced power consumption.

Peer Robotics

Jun 2020 – May 2022

Embedded Systems Engineer

- Re-designed and optimized the sensor integration system for real-time communication, reducing the number of components and communication ports, simplifying cabling, and enabling sensor response parameters to be tuned to system requirements, while scaling to multiple sensors with no added processing overhead.
- Developed **AVR based power distribution board's** firmware with custom drivers for all the peripherals. Executed real time operations by synchronizing with timers and interrupts. Scarcity of timers for different routines also led to the development and integration of a leapfrog mechanism, eliminating need for porting RTOS altogether.
- Developed VRU subsystem based on BNO080. The board involved a bridge (USB - I2C) micro-controller (STM32 - Cortex M0) to handle real-time responses and configuration of VRU. Other interfaces were provided for indicator controls.
- Developed **Health Care module** as part of **QDIC (Qualcomm Design in India Challenge)**. The module was built around SC66-E module and dealt with interfacing RFID sensor (over I2C) and IO.
- Developed **Human Robot Interaction** algorithm based on actuator's data enabling user to drive the robot manually by force without actually feeling the weight of the load on the robot and the robot itself.
- Designed and implemented communication and system's architecture for 250 kg payload product which included Siemens PLC, Modbus Protocol, CAN bus. Also, developed various drivers for supporting hardware like OPC UA, CAN, and OpenCAN.
- Designed and developed auto-charging system for the RM100 and RM250 (mobile robots). The project involved interface protocol development over Siemens S7 PLC and state execution in the robot.

- Played a critical role in deploying robots across warehouses and coordinating with on-ground personnel to ensure effective management.

PROJECTS

ARM NEON Optimization

Google Summer of Code'19

Feb 2019 – Dec 2019

dav1d, VideoLAN

- Achieved significantly faster execution of various functions compared to C implementation by using assembly language with SIMD architecture on ARM Cortex A53 and A73.
- Programmed crucial modules such as blend, inverse transformation matrix, and intra prediction, achieving a performance improvement of $\sim 10\times$ over the C implementation.
- Utilized optimization techniques including loop unrolling, memory alignment, and instr. reordering.
- The project resulted in extended runtime for decoding video on apps supporting dav1d on a single charge.

SDHCI MMC Driver

Google Summer of Code'18

May 2018 – Aug 2018

Haiku OS

- Implemented support for SDHCI controller according to SD Host controller spec. for Haiku OS on x86 arch.
- Project involved discovering devices on PCI bus and registering child nodes through different bus and device drivers. Controller registers were accessed through MMIO.
- Implemented driver compliant with SD specification 8.0. Finally, read/write functions were implemented in the kernel space. Message Signaled interrupts (MSI) were also implemented in the driver.

VOLUNTEERING

- **ACCL Cluster Management** – Administrator of a compute cluster of tens of machines, networked storage nodes, 100G switches, and VMs supporting different research projects. 2023 – Now
- **VDD '19** – Speaker at VideoLAN Dev Days (Tokyo), presenting SIMD optimization techniques for the ARM architecture. Dec 2019